

ACUTE EFFECTS OF ALMOND PROTEIN POWDER OR WHEY PROTEIN ON PLASMA AMINO ACID PROFILES IN POSTMENOPAUSAL WOMEN: A RANDOMISED CROSSOVER STUDY

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INTRODUCTION

Following menopause, women often experience a range of physiological changes such as increased visceral adiposity [1], accelerated loss of muscle mass [2] and metabolic dysregulation [3] due to the rapid decline of ovarian hormones. Consuming sufficient dietary protein may help attenuate the loss of muscle mass by stimulating muscle protein synthesis [4] while leading to favourable metabolic [5-6] and body composition outcomes [7]. Almond protein powder (APP) is a novel plant-based protein which has the potential of offering various health benefits for postmenopausal women beyond dietary protein, as its rich concentration of dietary fibre and monounsaturated fats may also lead to positive cardiometabolic health outcomes [8]. Currently, little is known about the amino acid, glycaemic and insulinaemic responses following APP ingestion in postmenopausal women. Thus, the aim of this study is to characterise the postprandial aminoacidaemia response of APP compared to whey protein (WP), placebo and water in postmenopausal women, with a secondary focus on assessing postprandial glucose, insulin and incretin responses.

METHODS

This randomised crossover study investigated the effects of four different treatments under the same conditions. Nine postmenopausal women aged 50-65 years attended four separate sessions, ingesting a different treatment drink (APP, WP, placebo or water) on each occasion. Venous blood samples were collected to determine postprandial aminoacidaemia responses (0-180 minutes). Linear mixed models (LMMs) were used for time-course analysis to allow for between-treatment comparisons of amino acids, glucose, insulin and incretin responses. Incremental area under the curve (iAUC) was calculated to compare the postprandial changes from baseline between treatment groups for amino acids, glucose, insulin and incretins.

RESULTS

A significantly greater iAUC was observed in the WP treatment compared to APP treatment for TAA, EAA, BCAA and leucine ($p<0.001$), while both APP and WP treatment groups showed significantly greater iAUC than the placebo and water treatment groups for TAA,

EAA, BCAA and leucine ($p<0.05$). No significant differences were observed in glucose iAUC between the four treatment groups. There were no differences in insulin iAUC between APP and WP ($p=0.684$). Time-course analysis revealed that leucine and BCAA concentrations were significantly higher in the WP group compared to APP, placebo and water groups at all time points except at baseline ($p<0.001$). Leucine and BCAA concentrations for APP were significantly greater than water and placebo treatment groups at 90 minutes ($p<0.001$) and 120 minutes ($p<0.05$). Glucose concentrations were not significantly different between APP and WP at all time points except 30 minutes ($p<0.002$). Insulin concentrations were greater in WP compared to APP only at 15 minutes ($p<0.001$) while the placebo treatment demonstrated significantly greater insulin concentrations than WP, APP and water treatments between 30-45 minutes ($p<0.05$). The WP treatment displayed a greater insulin concentration compared to APP and water at 15 minutes ($p<0.001$).

CONCLUSIONS

APP ingestion produced lower peak concentrations and excursions when compared to WP ingestion, which raises a question about APP's relative anabolic potential at this given dose. However, muscle protein synthesis was not directly measured, so one cannot conclude that APP yields a less anabolic response. This study demonstrated that APP is more effective in stimulating postprandial aminoacidaemia compared to water and placebo while producing similar glucose and insulin responses to that of WP. Further research is necessary to understand the role of APP in supporting skeletal muscle and cardiometabolic health in postmenopausal women and whether it is a viable plant-based alternative to whey protein.

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